

Design and development of IoT-based SCADA platform for OM in photovoltaic utility plants

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Abstract—This paper details the design and development of an Internet of Things (IoT)-based SCADA platform aimed at enhancing Operation and Maintenance (O&M) in photovoltaic (PV) power plants. Addressing the limitations of traditional SCADA systems, the proposed platform integrates technologies such as LoRa, WiFi, cloud computing, and data analytics. The architecture features an IoT gateway for versatile data acquisition and a cloud-based system for data storage, processing, and visualization. Robust security is implemented using VPN and X.509 certificates. The development followed a structured methodology including design, hardware/software creation, integration, and validation. Key results include an IoT gateway hardware and a complementary web-based monitoring interface. This platform provides a comprehensive solution for real-time monitoring and control, with the goal of improving the efficiency, reliability, and overall performance of O&M operations in large-scale PV installations.

Index Terms—Internet of Things, SCADA, photovoltaic systems

I. INTRODUCTION

The installed capacity of photovoltaic (PV) systems has been growing rapidly in recent years, with a significant increase in the number of utility-scale plants [1]. This growth has led to a greater need for effective monitoring and control systems to ensure optimal performance and reliability [1].

SCADA (Supervisory Control and Data Acquisition) systems play a crucial role in the operation and maintenance (O&M) of PV plants [2], providing real-time data acquisition,

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monitoring, and control capabilities [3], [4]. However, traditional SCADA systems often face limitations in terms of flexibility, scalability, and integration with modern communication technologies [3]. Using IoT (Internet of Things) technologies can enhance the capabilities of SCADA systems, enabling more efficient data collection, processing, and analysis, in addition to extending the system's reach to remote locations [1], [3].

In [5], a five-layer IoT-based SCADA system is presented, comprising perception, network, security, middleware, and application layers. An ESP32-E microcontroller acts as the Remote Terminal Unit (RTU), acquiring voltage and current data from PV modules and batteries while controlling load status based on battery levels. Data are transmitted to the Arduino Cloud via MQTT and to a Master Terminal Unit (MTU), implemented with an ESP32-S3, via HTTP. The MTU forwards data to the ThingSpeak platform for analytics and cloud storage, while also sending load images to a web server. The system demonstrated robustness and versatility, even during planned downtime.

In [6], a low-cost, open-source SCADA architecture is proposed for PV monitoring. The system employs an Arduino Uno as the RTU for data acquisition, a Raspberry Pi with Node-RED for communication, and an EmonCMS server for local data logging and visualization. Similarly, in [7], an Arduino Nano is used as the RTU, while the open-source Blynk platform functions as the MTU, enabling remote monitoring and control through mobile and web dashboards. The system supports both GSM and Wi-Fi communication, enhancing connectivity and accessibility.

In [2], a modular open-source SCADA architecture is proposed for PV plant monitoring. Multiple Raspberry Pi units are deployed, each dedicated to specific services such as com-

munication, application, and backup, improving system robustness. Since many photovoltaic plants already have sensors that do not integrate with the IoT concept, the integration of these legacy sensors is achieved through Modbus RS-485 converters, with Node-RED managing MQTT-based communication and implementing a security layer for broker access.

These studies illustrate significant progress and diverse approaches to SCADA integration in PV systems. However, only [2] addresses compatibility with legacy sensors, without supporting the simultaneous monitoring of multiple plants. In contrast, the proposed system enables both multi-plant supervision and interoperability with legacy equipment.

Furthermore, ensuring secure communication between the field devices, the IoT gateway, and the cloud platform is paramount, particularly given the vulnerabilities inherent in distributed IoT systems [3]. To address this critical aspect, the project implemented robust security measures. Secure communication channels were established using Virtual Private Network (VPN) technology, creating encrypted tunnels for data transmission between the IoT gateway and the central system. Furthermore, a comprehensive certificate management system, based on X.509 certificates, was employed to authenticate devices, encrypt data, and ensure the integrity of communications, mitigating risks such as unauthorized access and data tampering [3].

The system presented in this work was designed to function as a platform for O&M in PV utility plants, integrating custom algorithms and systems to improve the monitoring and control of PV plants. This paper details the design and development of this Industrial IoT-based SCADA platform [3], [8]. The proposed system integrates various IoT technologies, including wireless communication protocols (LoRa, Wi-Fi), cloud computing, and data analytics, to provide a comprehensive solution for real-time monitoring and control, working as a platform for custom manual and automatic operations, as well as algorithms applications [9]. The system architecture is described, along with the key components and functionalities.

The implementation of the IoT hardware and software is discussed, highlighting the challenges and solutions encountered during the development process. The paper also presents the results of the system validation, demonstrating its effectiveness in improving the monitoring and control of PV plants.

II. SYSTEM ARCHITECTURE

Fig. 1 illustrates the system architecture of the IoT-based SCADA platform for O&M in PV utility plants.

The system is composed by two main components:

- **Cloud-Based System**
- **IoT Gateway**

The cloud-based system complements the IoT gateway by providing a platform for data storage, processing, and visualization. It includes a database to manage the collected data, a backend for processing and evaluating information, and a frontend for user interaction and dashboard.

The frontend, implemented using React and Next.js, provides an interactive interface for real-time and historical vi-

sualization of sensor data, as well as alarm monitoring. The backend, implemented with Django and REST technologies, serves as the core processing unit. It handles data ingestion, storage, and retrieval, responding to frontend requests with filtered and structured outputs.

The IoT gateway serves as the central hub for data collection and communication. The board is equipped with a high-performance central processing unit (CPU) with 128 MB of volatile memory (RAM) and 32 MB of non-volatile memory. The gateway integrates three Ethernet ports, which can be used for communication with dataloggers, execution of commands on actuators, acquisition of inverter data, or transmission of information via the HTTP protocol to the SCADA system backend. Additionally, the board provides RS-485 interface to implement MODBUS protocol and USB interface, wireless communication via LoRa and Wi-Fi, and support for external storage through an SD card. These capabilities ensure high interoperability with devices commonly deployed in photovoltaic plants, including meteorological sensors and various inverter models.

In addition to HTTP-based communication, the IoT gateway supports data transmission via the MQTT protocol, in which it employs a publish-subscribe architecture and requires a broker to manage message routing, with the open source Mosquitto broker used for system testing. To forward messages from the IoT gateway to the backend through RESTful endpoints, an intermediary service, referred to as the MQTT Router, is employed. This service acts as a protocol bridge, enabling seamless integration between the MQTT publish-subscribe mechanism and the backend's HTTP-based interface.

The developed hardware enables reliable data acquisition from various sensors and equipment in the photovoltaic plant, including inverters and dataloggers, through the MODBUS protocol over TCP/IP or RS-485 interfaces. Moreover, the IoT gateway integrates a wireless sensor network with LoRa communication, providing greater flexibility in the deployment and distribution of sensors across large-scale solar facilities. Both the system architecture and the IoT gateway design were conceived to ensure compatibility with devices commonly installed in photovoltaic plants, thereby minimizing the need for additional investments in new infrastructure and enabling the efficient and scalable implementation of an IoT-based SCADA system.

To ensure secure communication between the IoT gateway and the cloud-based system, a Virtual Private Network (VPN) was implemented. This creates an encrypted tunnel, safeguarding data integrity and confidentiality during transmission over public networks. Furthermore, communication with remote sensors via LoRa utilizes a custom-developed protocol incorporating cryptographic capabilities. This ensures that data exchanged between the gateway and the sensors is protected against unauthorized access and tampering, enhancing the overall security posture of the distributed system.

III. METHODOLOGY

The methodology for developing the IoT-based SCADA platform involves several key steps:

- **System Design:** The system architecture is designed to integrate various components, including the IoT gateway, cloud-based system, and field devices. Communication protocols and data flow are defined to ensure seamless integration.
- **Hardware Development:** The IoT gateway hardware is developed, incorporating multiple wireless communication technologies. The hardware is designed to interface with various field devices, such as inverters, solarimetric stations, and PLCs.
- **Software Development:** The embedded software for the IoT gateway is developed, including firmware for microcontroller-based LoRa communication and application software for embedded Linux. The cloud-based system is also developed, including the database, backend, and frontend components.
- **System Integration:** The hardware and software components are integrated to form a cohesive system. Data acquisition and transmission are tested to ensure reliable communication between the IoT gateway and cloud-based system.
- **Validation:** The system is validated using simulated data to ensure its robustness and reliability. Key functionalities are tested to evaluate the system's performance in real-world scenarios.
- **Deployment:** The system is deployed in a PV utility plant for real-time monitoring and control. The system's performance is continuously monitored, and improvements are made based on user feedback and operational requirements.

IV. RESULTS

The hardware integrates multiple wireless communication technologies, including LoRa, WiFi, Ethernet, and RS-485, enabling robust and flexible connectivity. The embedded software developed for this hardware consists of two main components: a firmware running on a microcontroller to handle LoRa communication and an application software designed for embedded Linux and a custom operating system, as shown in Fig. 2. This combination ensures seamless data acquisition and transmission from various field devices.

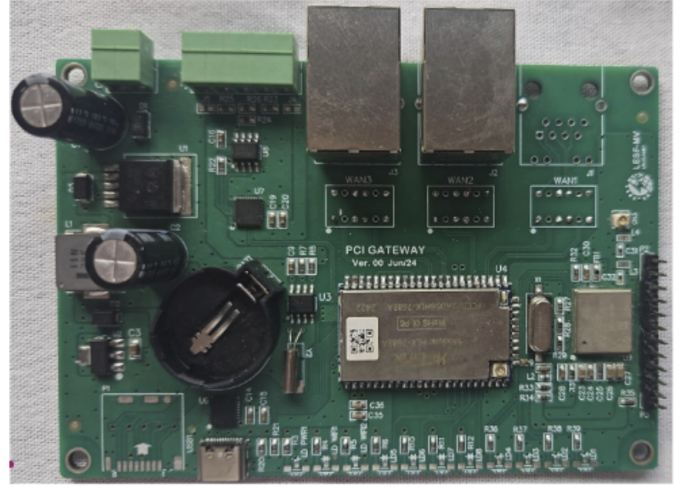


Fig. 2: Developed IoT Gateway hardware.

Additionally, a web-based system was developed to complement the IoT gateway. As described in Section II, the backend processes incoming data from the IoT gateway, evaluates the information, and triggers alarms when necessary. The frontend provides an intuitive interface for users to monitor system

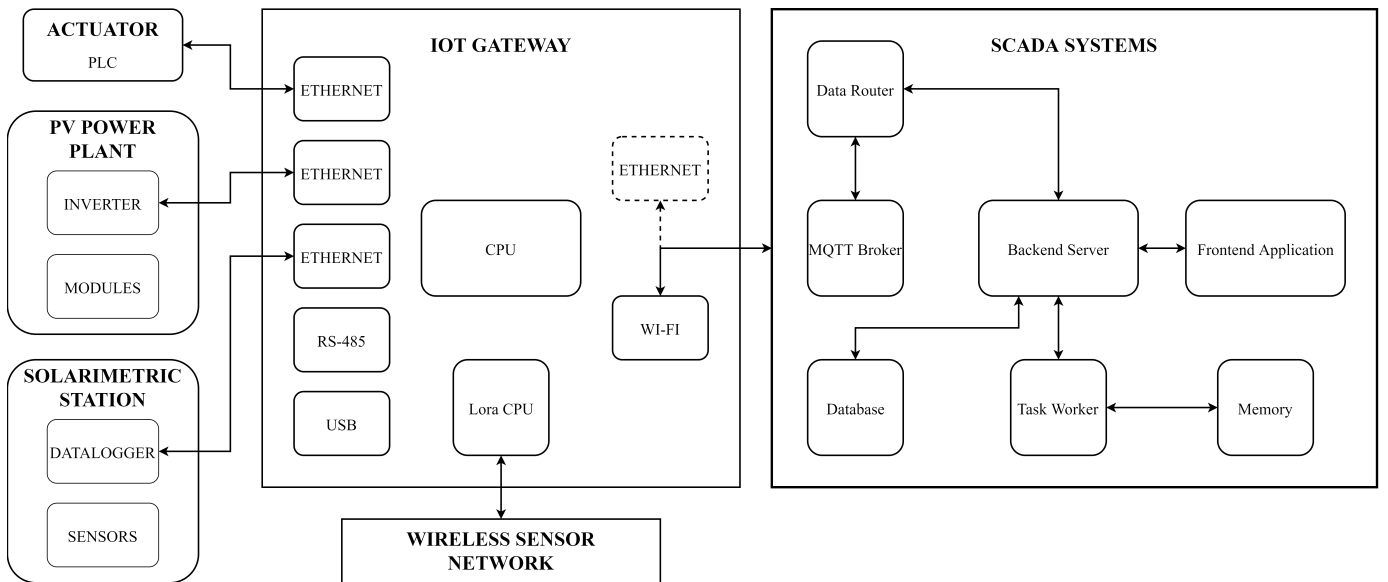


Fig. 1: System architecture of the IoT-based SCADA platform for O&M in PV utility plants.

performance, visualize relevant outputs, and respond to alerts effectively. Together, these components form a cohesive solution for real-time monitoring and control in photovoltaic utility plants. In the web system the user is capable of monitoring multiple pv plants.

A. System testing and validation

1) *Monitoring and Data Acquisition System of the Pilot Plant:* Instrumentation and sensor installation are fundamental steps in monitoring photovoltaic plants. To test and validate the developed IoT-based SCADA platform, a variety of typical sensors were installed. Data acquisition was performed using two distinct approaches: a traditional system with a commercial datalogger and a proprietary system based on an IoT-integrated Wireless Sensor Network (WSN) [10].

2) *Datalogger Integration and Data Validation in the IoT-based SCADA platform:* The tests were conducted at the pilot photovoltaic plant of the Universidade Estadual de Campinas (Unicamp), as illustrated in Fig. 3. On-site, a commercial datalogger was installed to integrate the sensors, including irradiance sensors (reference cells and pyranometers), a thermo-hygrometer, an anemometer, and module temperature sensors (PT100).



(a) PV module string with tracking system.



(b) On the right, the solarimetric station with a commercial datalogger (in red), a thermo-hygrometer, and an anemometer.

Fig. 3: Pilot photovoltaic plant at the University.

In Fig. 4, sensors connected to the datalogger are presented, which was connected via Ethernet to the local server hosting the IoT monitoring system.



Fig. 4: Sensors installed at the pilot plant: reference cell (right) and pyranometer (left).

Data from the sensors, centralized by the datalogger, is displayed and stored by the monitoring system. This information is crucial for diagnosing the plant's performance and for decision-making regarding both preventive and corrective maintenance. In addition to being essential for efficiency analysis, the data is also used to train algorithms for fault and anomaly detection. To this end, the monitoring system already integrates Machine Learning algorithms to perform this task automatically [11].

Fig. 5 illustrates the variation of Global Horizontal Irradiance (GHI), in W/m^2 , recorded over four days at the pilot plant. The data were collected by a pyranometer installed at the solarimetric station. These measurements are stored within the monitoring system, where they can be retrieved for analysis and serve as input to plant performance evaluation algorithms. Additionally, when the plant's power generation deviates from expected values for the local meteorological conditions, the system generates warnings.

3) *Testing and validation of the IoT-integrated Wireless Sensor Network connected to the monitoring system platform:* In large-scale plants, systems employing dataloggers are extremely common [12] [13] [14]. Thus, it is essential that the proposed monitoring system be ready to receive data from dataloggers already existing at the pv plants. However, for smaller plants where resources are more limited, dataloggers and complete solarimetric stations are less common, as the cost of installation and maintenance of this equipment is high [15]. It is at this point that the need for the monitoring system to include an IoT wireless sensor network becomes more evident, which is why the integration of the IoT system proposed in [10] is so relevant in the current context.

The data acquisition hardware presented in [10] was used to collect data from the PT100 temperature sensor, which is shown in Fig. 6. In this setup, the IoT network's sensor node reads the temperature and sends the data to a concentrator node.

This concentrator acts as an interim IoT gateway while the final hardware (presented in Fig. 2) is under development. The concentrator node connects to the local server of the



Fig. 5: Variation of Global Horizontal Irradiance (GHI) measured by a pyranometer at the pilot plant over four days.

monitoring system, allowing the platform to collect, display, and store the data. The firmware running on this concentrator corresponds to the first version developed for the proposed IoT gateway Lora communication.



Fig. 6: Placement of the PT100 temperature sensor for the data acquisition validation test.

Temperature data from the IoT sensor network was integrated into the monitoring system and made available on a dedicated interface. The data was collected over several days

at the monitoring station of the pilot plant in Unicamp, and the results for 12 days of operation can be seen in Fig. 7.

The figure demonstrates the system's ability to continuously and efficiently collect, store, and display data from the IoT network. In the long term, the plan is to expand the network to include other sensors relevant for the photovoltaic plant's efficiency measurements, integrating them into the IoT-based SCADA system.

V. CONCLUSION

The development of a SCADA system with integrated IoT features has demonstrated significant progress, addressing the challenges of real-time data acquisition and intelligent processing for solarimetric and inverter systems. The core system successfully collects meteorological data and sensor readings, enabling the foundation for advanced dashboard visualization and operational control.

Key functionalities are designed to enhance the system's intelligence and reduce unnecessary alerts, ensuring a more efficient monitoring and operational experience. These features operate dynamically, processing data upon arrival from remote sensors to provide timely and actionable insights.

In conclusion, the SCADA system with IoT integration presents a significant step forward in intelligent monitoring and control for renewable energy systems. Thereby, the proposed IoT-based SCADA system integrates legacy devices, wireless sensor networks, and cloud computing into a unified platform, enhancing scalability and real-time monitoring in PV utility plants.

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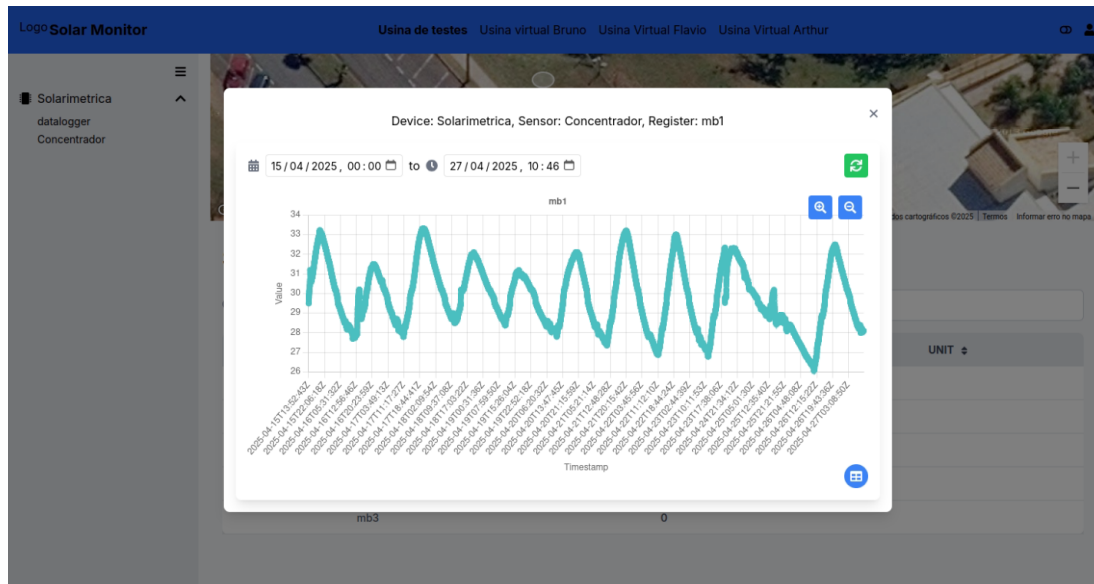


Fig. 7: Concentrator integration test: gateway firmware installed on a data acquisition board and integrated into the monitoring system. Temperature ($^{\circ}\text{C}$) reading from the PT100 sensor at the pilot PV plant monitoring station. Data not associated with inverter operation, as the inverter was not yet turned on.

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